

RADP — choose the parser by **retrieval, not by appearance; train only where it helps**

PROBLEM C1 · The disconnect

Parser output

Appearance

BC · edit-dist

MinerU ranks #1

Retrieval

RCPS

MinerU ranks LAST

BC ↔ RCPS $r = -0.81$ · parser choice → Hit@1 **2.8x**
(0.20→0.55)

DIAGNOSE C2 · Coverage diagnostic

gold answer span

in parser output?

fits one chunk?

SPLIT ≤2.3% chunker fault

ABSENT 20.2% parser fault

COVERED

absent constant across 8 chunkers → fix the parser, not the chunker

SELECT C3 · RCPS protocol

held-out Q-A probe + parsed corpus

chunk → retrieve (MRR)

✓ **extrinsic**
(probe)

✓ **retriever-avg**
x3

✓ **format-invariant**

RCPS ranking → pick parser + chunker (no training)

retriever-avg flips the top parser a single embedder gets wrong ($\tau = 0.87$)

IMPROVE C4 · Bounded training

Prod
parser

sample K
parses

rank
candidates

LoRA-toggle
DPO

rank by: edit-dist → GT (RADP-Distill ✓)
vs page-RCPS (RADP-DPO) — same result

retrieval reward unnecessary → distil to clean GT text
+1.22 pp Hit@5 (OHR-Bench)